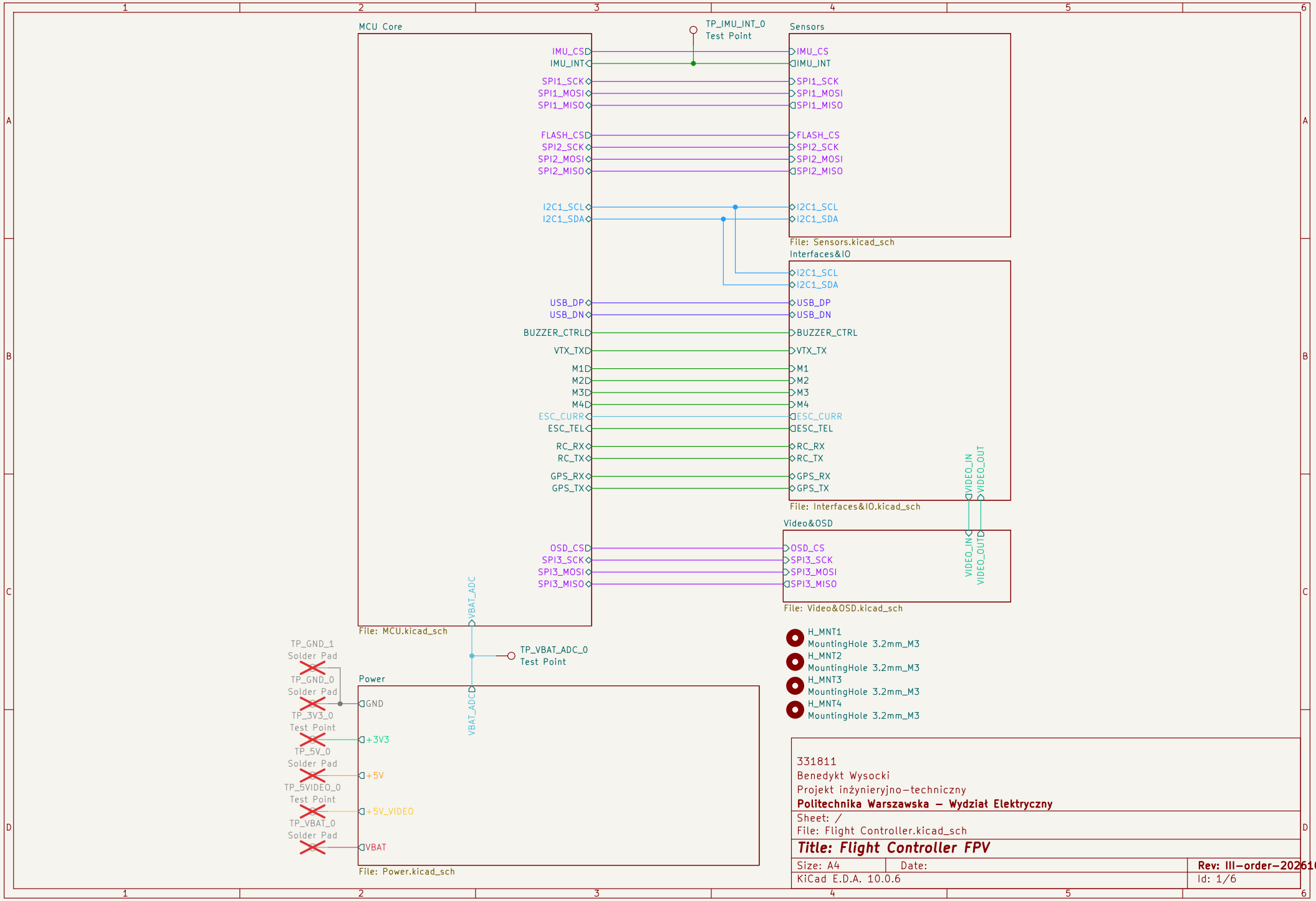
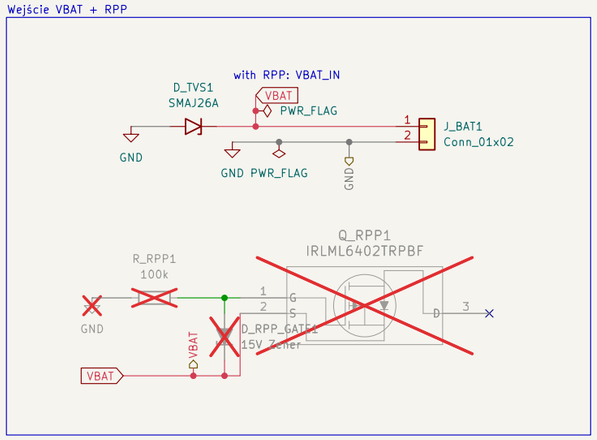
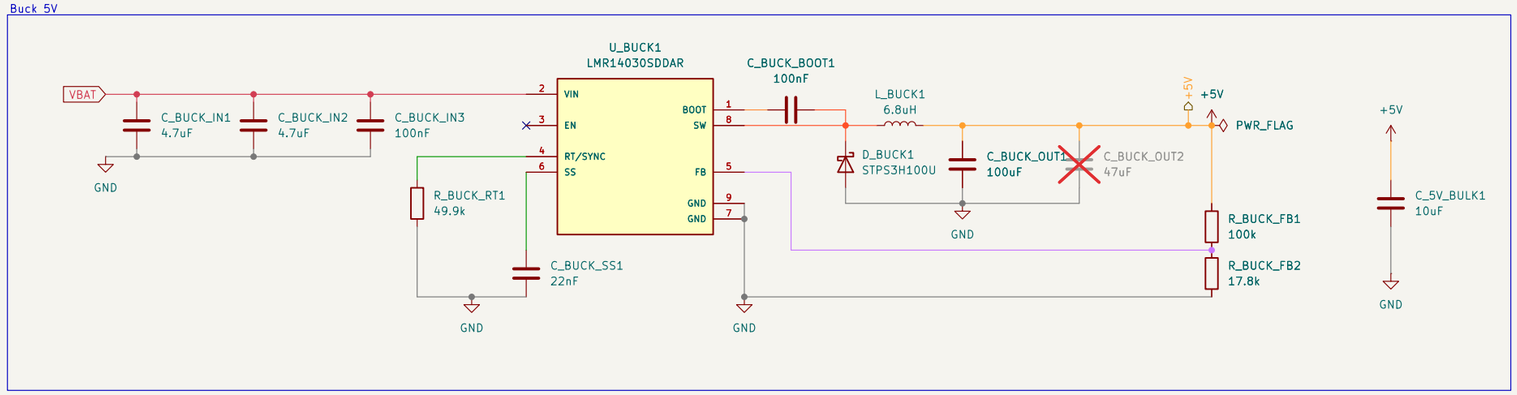

MCU Core
IMU_CS
IMU_INT
SPI1_SCK
SPI1_MOSI
SPI1_MISO
FLASH_CS
SPI2_SCK
SPI2_MOSI
SPI2_MISO
I2C1_SCL
I2C1_SDA
USB_DP
USB_DN
BUZZER_CTRL
VTX_TX
M1
M2
M3
M4
ESC_CURR
ESC_TEL
RC_RX
RC_TX
GPS_RX
GPS_TX
OSD_CS
SPI3_SCK
SPI3_MOSI
SPI3_MISO
VBAT_ADC
File: MCU.kicad_sch
TP_IMU_INT_0
Test Point
Sensors
IMU_CS
IMU_INT
SPI1_SCK
SPI1_MOSI
SPI1_MISO
FLASH_CS
SPI2_SCK
SPI2_MOSI
SPI2_MISO
I2C1_SCL
I2C1_SDA
File: Sensors.kicad_sch
Interfaces&IO
I2C1_SCL
I2C1_SDA
USB_DP
USB_DN
BUZZER_CTRL
VTX_TX
M1
M2
M3
M4
ESC_CURR
ESC_TEL
RC_RX
RC_TX
GPS_RX
GPS_TX
VIDEO_IN
VIDEO_OUT
File: Interfaces&IO.kicad_sch
Video&OSD
OSD_CS
SPI3_SCK
SPI3_MOSI
SPI3_MISO
VIDEO_IN
VIDEO_OUT
File: Video&OSD.kicad_sch
TP_VBAT_ADC_0
Test Point
TP_GND_1
Solder Pad
TP_GND_0
Solder Pad
TP_3V3_0
Test Point
TP_5V_0
Solder Pad
TP_5VIDEO_0
Test Point
TP_VBAT_0
Solder Pad
Power
GND
+3V3
+5V
+5V_VIDEO
VBAT
File: Power.kicad_sch
H_MNT1
MountingHole 3.2mm_M3
H_MNT2
MountingHole 3.2mm_M3
H_MNT3
MountingHole 3.2mm_M3
H_MNT4
MountingHole 3.2mm_M3
331811
Benedykt Wysocki
Projekt inżynieryjno-techniczny
Politechnika Warszawska – Wydział Elektryczny
Sheet: /
File: Flight Controller.kicad_sch
Title: Flight Controller FPV
Size: A4
Date:
Rev: III-order-20261003
KiCad E.D.A. 10.0.6
Id: 1/6

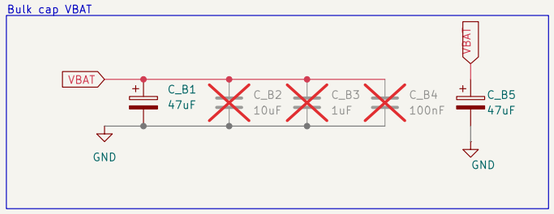
Wejście VBAT + RPP



Buck 5V

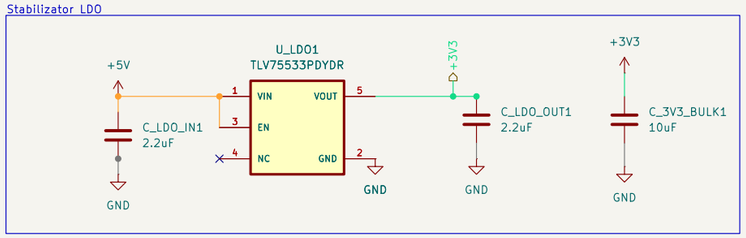


Bulk cap VBAT

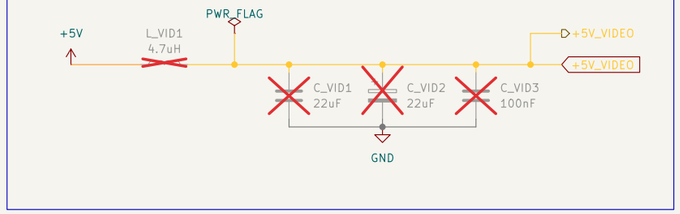


Sparrow III: C_B1 + C_B5 = 94uF / 50V, h=5.8mm. ESR pary ~340mR.

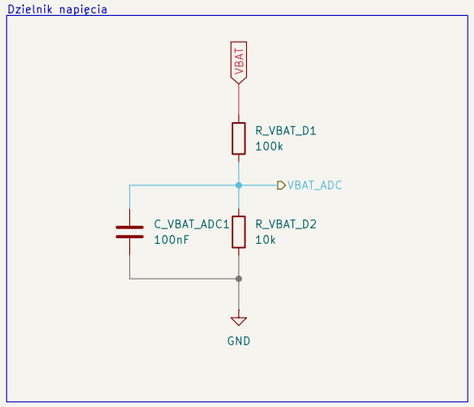
Stabilizator LDO



Zasilanie do kamery i nadajnika wideo (VTX)

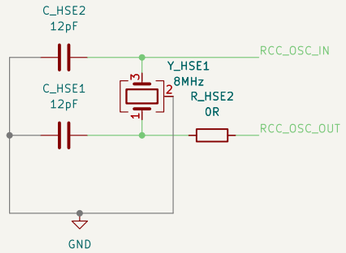


Dzielnik napięcia

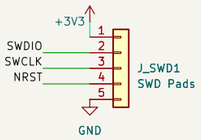


Sheet: /Power/		
File: Power.kicad_sch		
Title: Sparrow III – Flight Controller		
Size: A3	Date: 2026-09-10	**Rev: III-placement**
KiCad E.D.A. 10.0.6		Id: 2/6

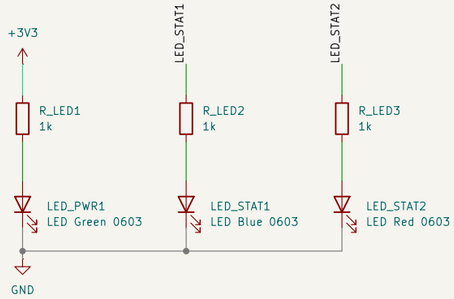
Oscylator kwarcowy



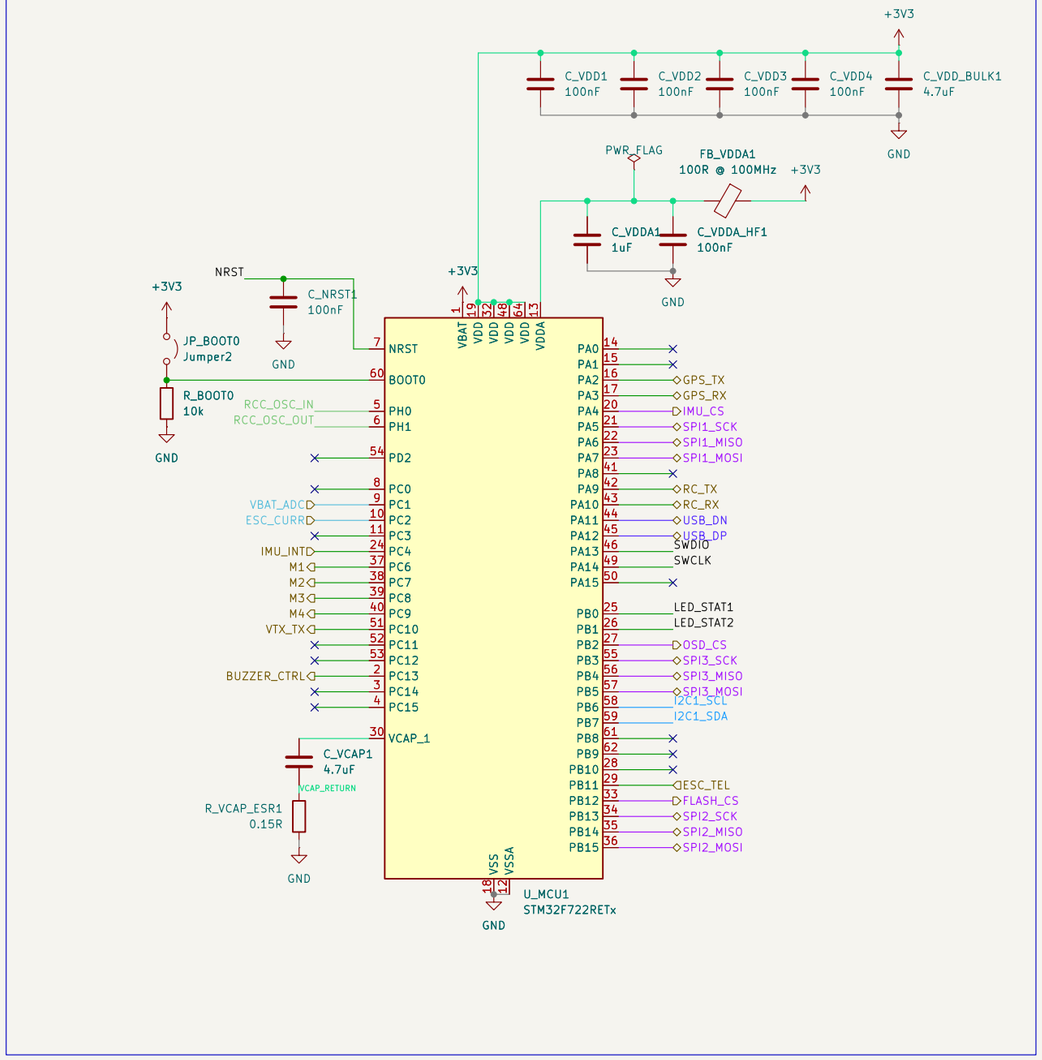
Serial Wire Debug



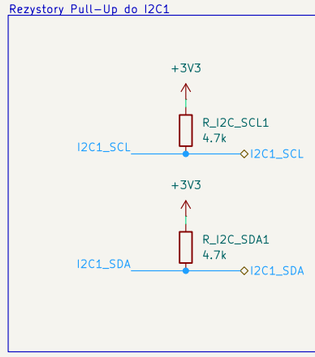
Diody diagnostyczne



MCU

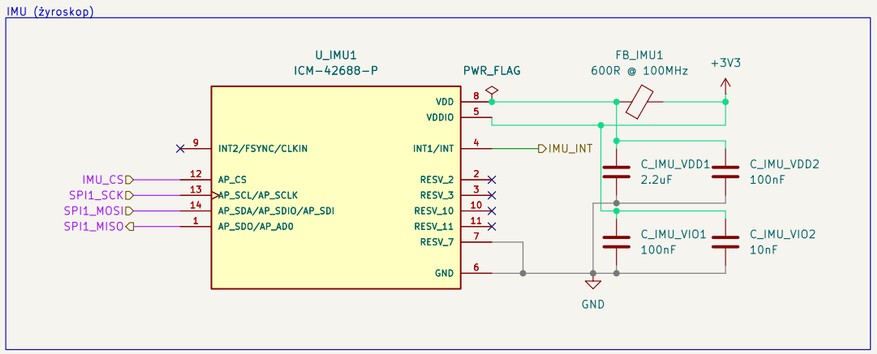


Rezystory Pull-Up do I2C1

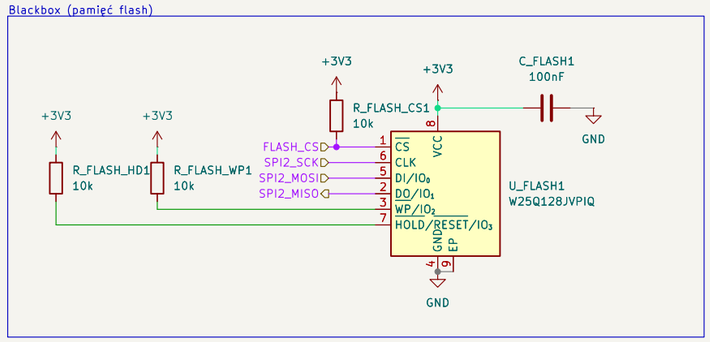


Sheet: /MCU Core/		
File: MCU.kicad_sch		
Title: Sparrow III — Flight Controller		
Size: A3	Date: 2026-09-10	**Rev: III-placement**
KiCad E.D.A. 10.0.6		Id: 3/6

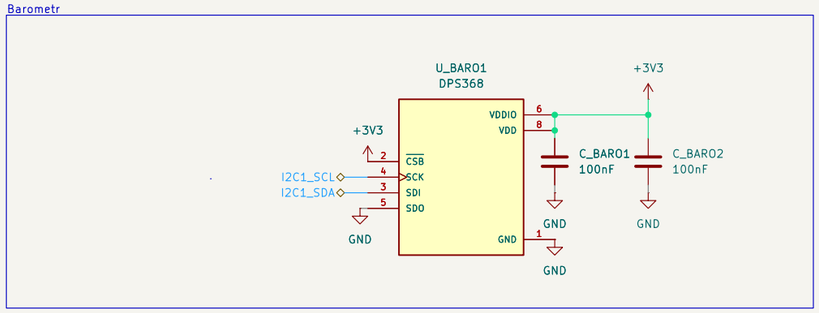
IMU (żyroskop)



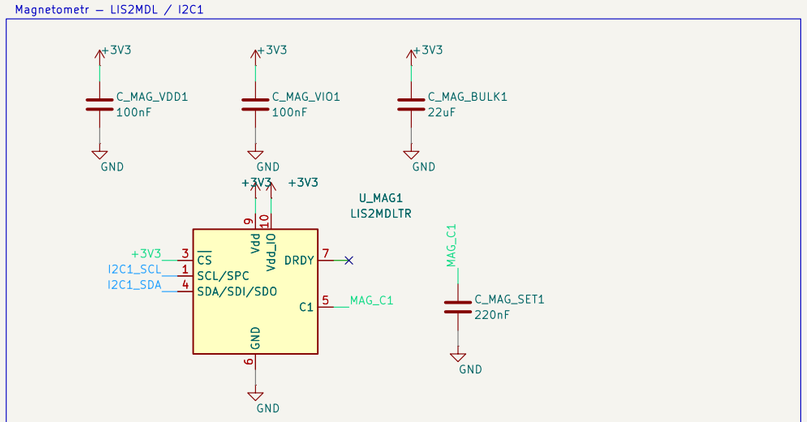
Blackbox (pamięć flash)



Barometr



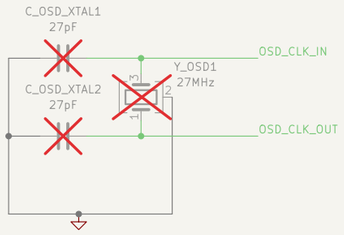
Magnetometr — LIS2MDL / I2C1



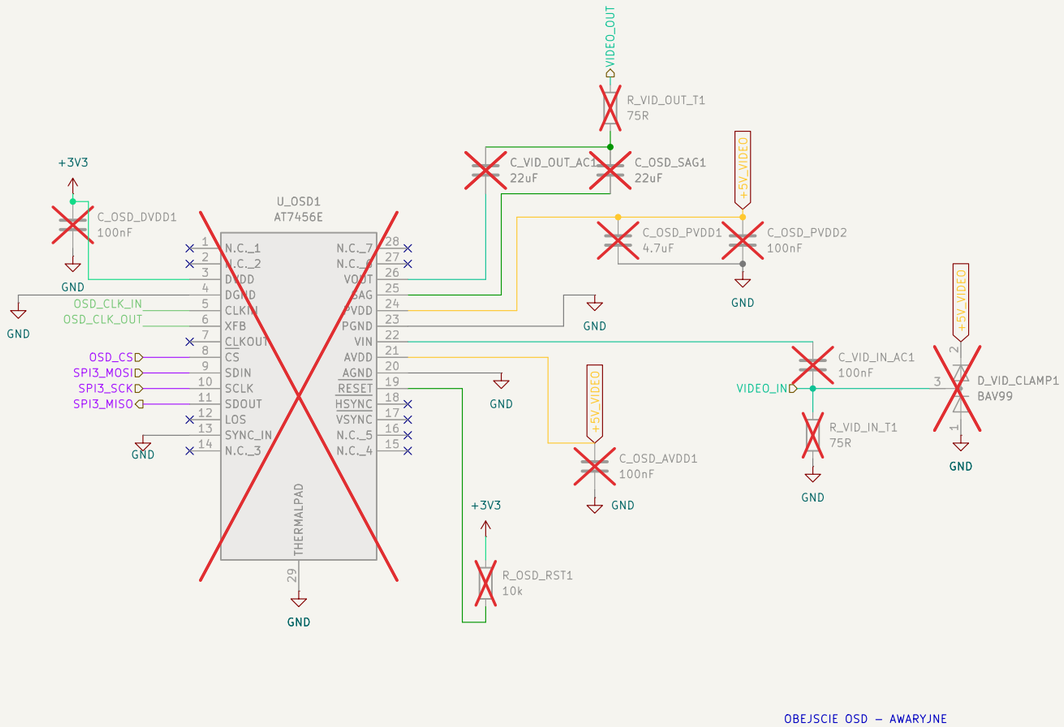
Sparrow III: LIS2MDL, I2C1 @ 0x1E. C1=220nF low ESR <=200mR.
GPS: choose compass with address other than 0x1E, or leave GPS I2C unconnected.
LGA requires controlled reflow / hot air; do not attempt iron-only assembly.

Sheet: /Sensors/	
File: Sensors.kicad_sch	
Title: Sparrow III — Flight Controller	
Size: A3	Date: 2026-09-10
KiCad E.D.A. 10.0.6	Id: 4/6
	Rev: III-placement

Oscylator kwarcowy do OSD



On-Screen Display (nakładka graficzna)

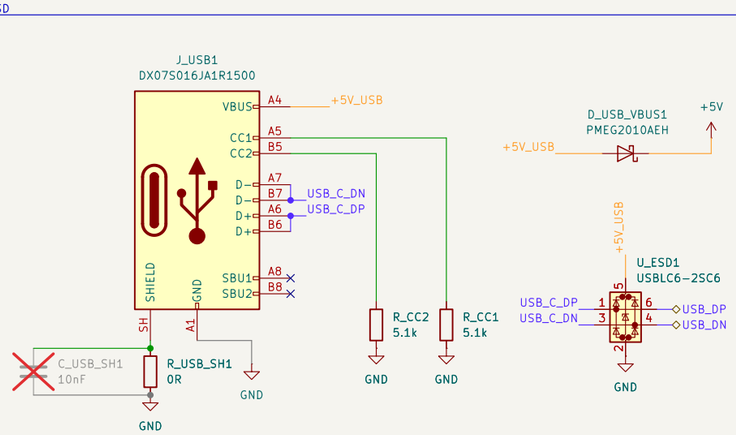


OBEJSCIE OSD – AWARYJNE

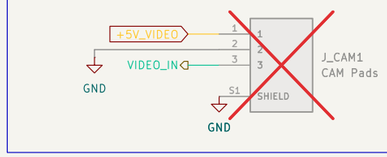


Sheet: /Video&OSD/		
File: Video&OSD.kicad_sch		
Title: Sparrow III – Flight Controller		
Size: A3	Date: 2026-09-10	**Rev: III-placement**
KiCad E.D.A. 10.0.6		Id: 5/6

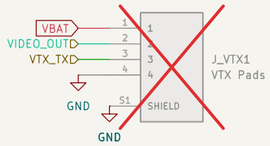
USB&ESD



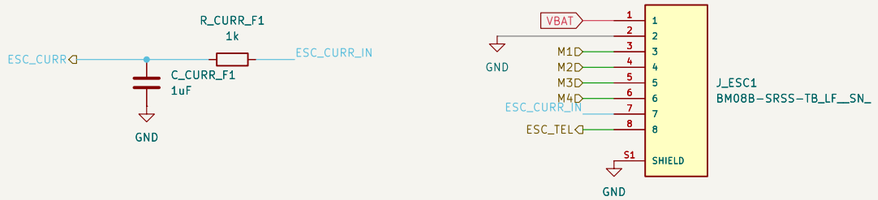
Kamera FPV



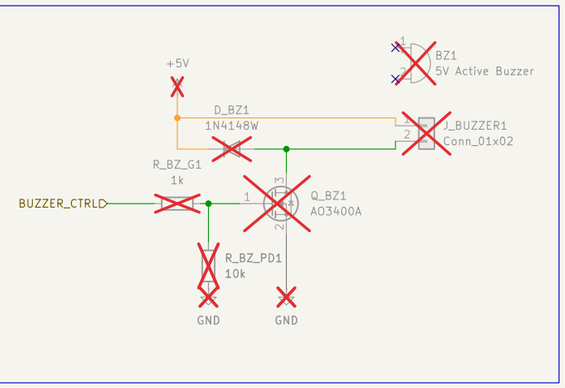
Nadajnik wideo



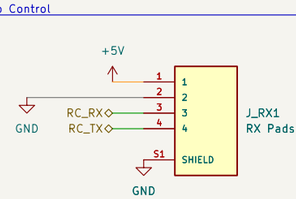
Magistrala do ESC + filtr RC dla pomiaru prądu



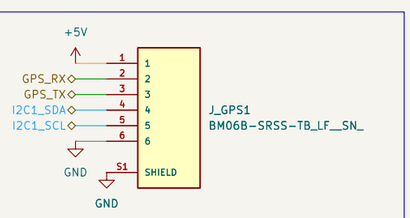
Buzzer



Odbiornik Radio Control



Złącze GPS



VTX – Sparrow III



Sparrow III: J_VTX2 = +5V / GND / SmartAudio (PC10).
J_VTX1 and J_CAM1 retained as DNP, excluded from PCB.

Sheet: /Interfaces&IO/		
File: Interfaces&IO.kicad_sch		
Title: Sparrow III – Flight Controller		
Size: A3	Date: 2026-09-10	**Rev: III-placement**
KiCad E.D.A. 10.0.6		Id: 6/6